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MESOZOIC PALAEOMAGNETISM OF THE TRANSDANUBIAN CENTRAL MOUNTAINS AND ITS TECTONIC IMPLICATIONS

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ABSTRACT

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The Mesozoic apparent polar wandering (APW) of the Transdanubian Central Mountains, determined from thermally isolated natural remanences at 13 localities, shows a remarkable similarity to the Mesozoic APW of Africa in that they both exhibit the same loop-like movement. Moreover, the difference between the two APW's can practically be eliminated by a 35° clockwise rotation of the palaeodeclinations. It is concluded, therefore, that the region of the Transdanubian Central Mountains was part of the African (-Adriatic) plate up to some time in the Cenozoic when it moved to its present position, resulting in a 35° anticlockwise rotation relative to Africa.

INTRODUCTION

Jurassic declinations from the Transdanubian Central Mountains were found to be rotated by about 70° counterclockwise relative to stable European palaeodeclinations (Márton and Márton, 1978). Since the observed palaeoinclinations did not differ significantly either from the European or the African ones computed for the region, the declination rotation was explicable either: (1) by a c. 70° westerly rotation of the Transdanubian Central Mountains relative to Europe after the Jurassic; or (2) by considering the Transdanubian Central Mountains as a quasi-autochthonous unit of the African plate which, however, was rotated to some degree to the West relative to Africa. The latter explanation was supported for the lower Jurassic, as the fauna is of Mediterranean type and the facies is ammonitico rosso and hierlatz limestone, both characteristic of the southern margin of the Tethys (Géczy, 1973). If the Transdanubian Central Mountains had been an African

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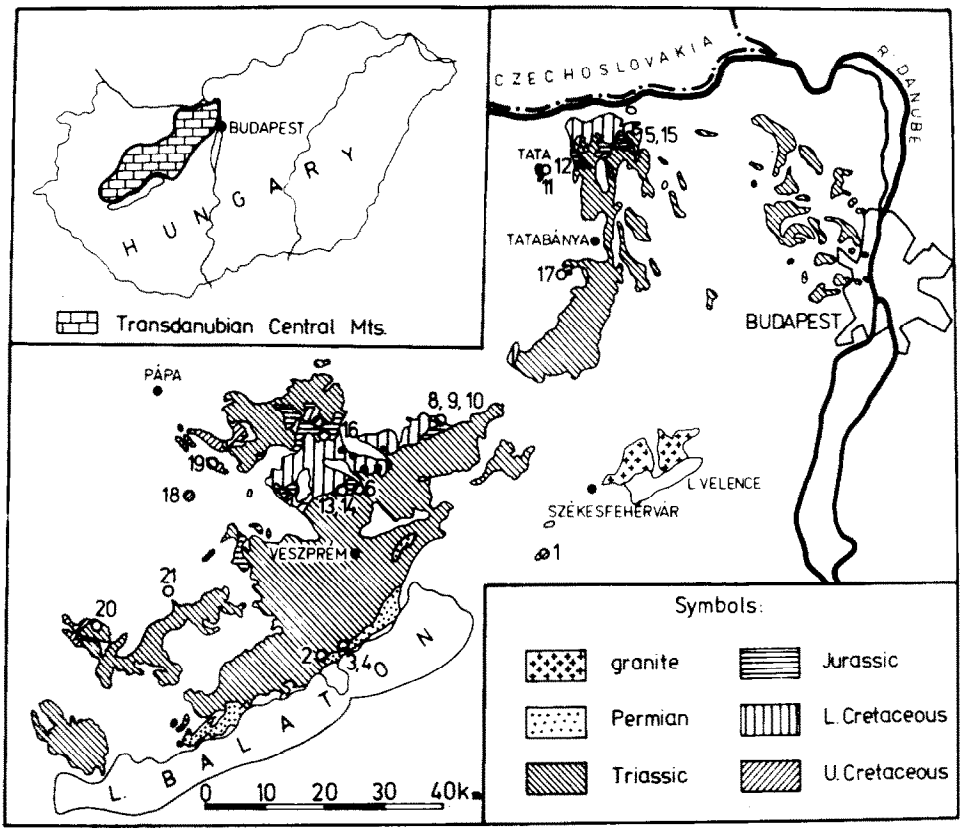


Fig. 1. Simplified geological map of the Transdanubian Central Mountains after Dank and Fülöp (1967) with the sampling localities (circled and numbered). For locality names refer to text.

tectonic unit, then they must have moved with Africa relative to Europe and this movement should be detectable by palaeomagnetic measurements. In order to test this explanation, palaeomagnetic investigations have now been extended to sedimentary rocks of other ages.

The Transdanubian Central Mountains comprise a complete sequence of Mesozoic marine sediments, which are believed to have formed in a wide epeirogenic depression and were subsequently uplifted and deformed, predominantly by fault-tectonics (Vadász, 1960). They rise a few hundred metres above sea level and are surrounded by fault-bounded younger depressions. The different ranges along the southwest—northeast strike are also called mountains and our sampling localities are situated in the Bakony, Vértes and Gerecse Mountains, respectively. Figure 1 shows a simplified geological map (Dank and Fülöp, 1967) with the sampling localities circles and numbered (cf. next section and Table I). The rocks collected are red, white and grey limestones and marls.

The magnetic measurements were done on a JR-4 spinner magnetometer with an rms noise level of 4×10^{-6} Am^{-1}. All samples were thermally cleaned in a Schonstedt TSD-1 unit at temperatures determined from progressive demagnetization of pilot specimens. The relevant magnetic mineralogy was studied using remanent coercivity spectra derived from IRM aquisition characteristics (Dunlop, 1972). Magnetite and haematite were distinguished on the basis of their remanent coercivities, magnetite being the softer component, and the highest coercivity fraction was identified with pigmentary haematite. The thermal demagnetization characteristics and the remanent coercivity spectra together served as a means to infer whether the magnetization could be associated with the stratigraphic age of the rock.

SAMPLING AND ANALYSIS OF MAGNETIZATION

Devonian locality: 12 samples were collected from the Polgárdi quarry (*1*) comprising a crystalline white limestone with a pinkish tint. The rock is estimated to have been metamorphosed in the lower range of the greenschist facies during upper Devonian (?) times. Seven of the samples were characterized by haematite and only had a consistent magnetic direction at 400°C ($\bar{D} = 108.8°$; $\bar{I} = -30.9°$; $K = 54$; $\alpha_{95} = 8.3°$). Five other samples had magnetite plus haematite pigment (Fig. 2a) with a stable magnetization isolated at 400°C ($\bar{D} = 263°$; $\bar{I} = -19.6°$; $K = 22$; $\alpha_{95} = 16.8°$; $\Phi = -12.1°$; $\Lambda = 286.2°$; $\Delta p = 9.2°$; $\Delta m = 17.6°$) which persisted till the Curie-point of the magnetite.

Triassic localities: These were sampled at six places from rocks ranging

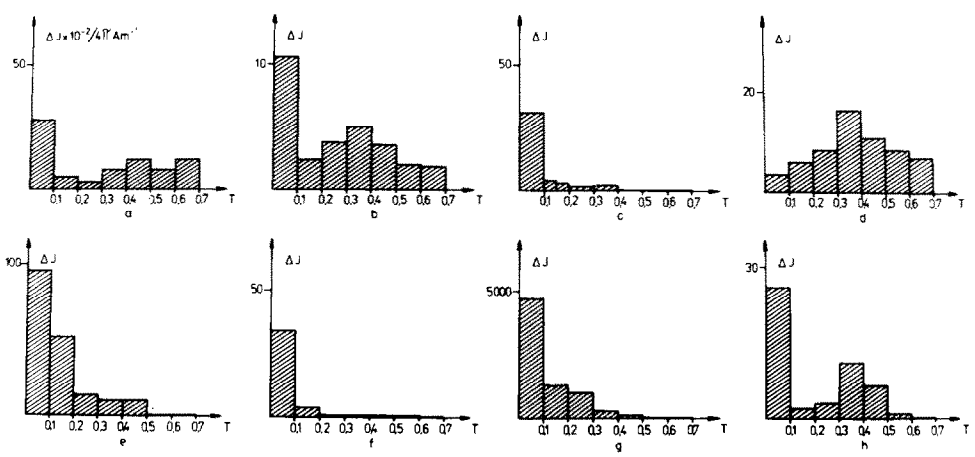


Fig. 2. Characteristic remanent coercivity spectra derived from isothermal remanences induced in magnetic fields up to 0.7 T for eight localities of the Transdanubian Central Mts. These are as follows: a, Polgárdi (*1*); b, Szákahegy (*2*); c, Nosztory valley (*3*); d, Borzavár (*7*); e, Bakonycsernye (*8*); f, Lókút-Hill (*14*); g, Berzsekhegy (*15*); h, Borzavár (*16*).

from lower to upermost Triassic in age from the Balaton Highlands, the Bakony and the Gerecse Mountains.

Száka-hegy quarry (*2*) near Balatonfüred town comprises gently dipping grey limestone beds of Seisian—Werfenian age. The magnetic minerals are magnetite and pigmentary haematite (Fig. 2b). Thermal demagnetization of the pilot specimen revealed both the importance of the haematite as the carrier of a secondary remanence and the feasibility of the isolation of another remanence component residing in the magnetite. These are demonstrated in Fig. 3a. The demagnetization path of the pilot specimen shows that the large normal overprint (carried by haematite) is stable up to 400° C but completely disappears after demagnetization at 500° C. Note that the stable endpoints of magnetizations (remanence directions above 500° C) for the normal samples are, on average, exactly opposite to the magnetic direction of the only reversed pilot specimen which also proves that cleaning is complete at 500° C.

Grey marls of middle Triassic age (upper Ladinian—lower Carnian) were collected from two neighbouring quarries on either side of the Csopak—Veszprém road some 2 km away from Csopak village.

The remanence coercivity spectrum in Fig. 2c is characteristic of the samples from the lower beds [left side quarry (*3*)] and shows the predominance of magnetite and no haematite pigment. On thermal demagnetization the stable remanence direction was revealed above 300° C and it was only destroyed at the Curie-point of magnetite.

The samples from the upper beds [right side quarry (*4*)] were very weakly magnetized with an initial remanence intensity of about 2×10^{-5} Am^{-1}. The remanence coercivity spectrum, though similar to the one in Fig. 2b, indi-

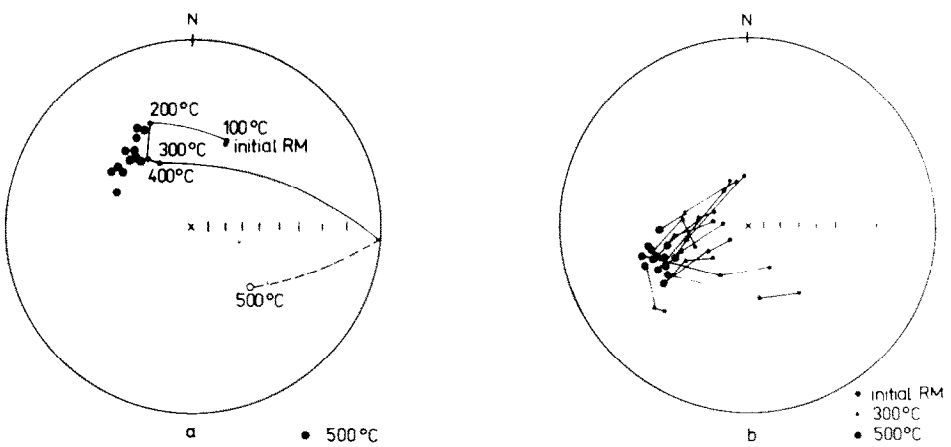


Fig. 3. Stereographic plots of remanence directions during, respectively after thermal demagnetization. a and b are for the samples from Szákahegy (*2*) and Lókút-Hill (*14*), respectively.

cated less magnetite. Thermal cleaning was only possible at 200° C because above 300° C the remanence intensities dropped below the noise level of the magnetometer. Even above 200° C the accuracy of the remanence measurements was poor and resulted, after converting the two reversed directions into normal ones, in a large α_{95} (cf. Table I).

From the uppermost Triassic (Rhaetian) limestone localities Kecskekö quarry (*5*) near Lábatlan (Gerecse Mountains) and Eplény quarry (*6*) near Veszprém (Bakony Mountains) did not yield results because the initial remanence directions were scattered, respectively aligned in the present field direction and the remanence intensities became too weak for measurement above 200° C, respectively 300° C demagnetization, without any sign of the directions getting cleaner.

The same material from the Borzavár village quarry (*7*) (Bakony Mts) was also of very low intensity except for two samples with a pinkish tint for which the remanent coercivity spectrum (Fig. 2d) indicated pigmentary heamatite alone. The (stable) remanence component whose direction (uncorrected for tilt $\bar{D} = 307.7°$; $\bar{I} = 42°$; $K = 772$; $\alpha_{95} = 9°$) was computed using four (2 + 2) specimens after demagnetization at 400° C, though most probably diagenetic, is consistent with other Mesozoic results from the Central Mountains.

Jurassic localities: Two sections were sampled both at Bakonycsernye (*8*,

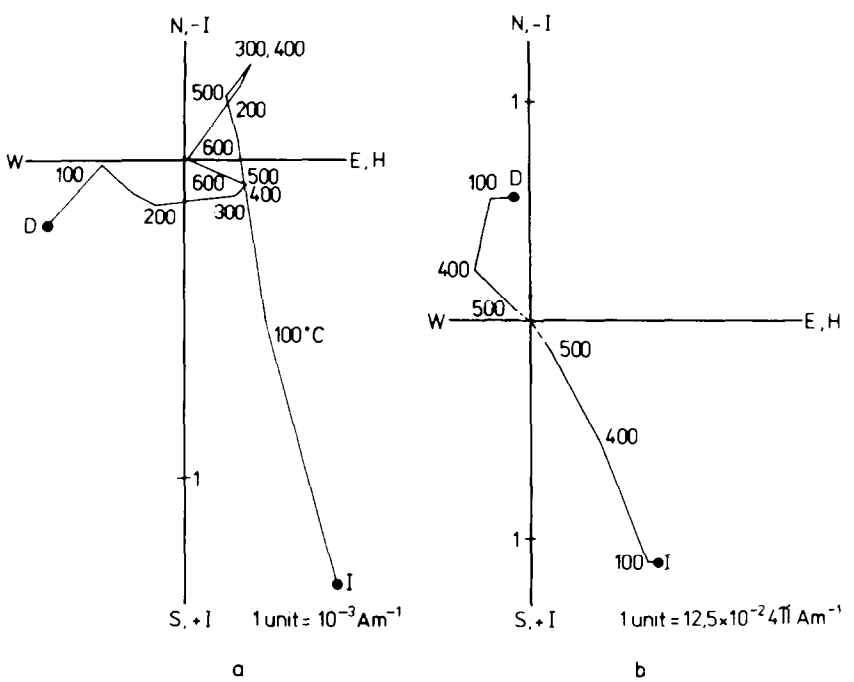


Fig. 4. Orthogonal thermal demagnetization plots. a is for a Jurassic red limestone specimen of reversed polarity (*8*) and b is for a Cretaceous dark-grey marl specimen (*15*).

9, *10*) and Lókut (*13*, *14*) in the Bakony Mountains and one each from the Kálvária-Hill at Tata (*11*) and Tardosbánya (*12*) in the Gerecse Mountains, for wider areal coverage. The rocks collected are red limestones of lower Jurassic age and only the Tithonian is represented by a white limestone at Lókut.

All red limestones possess very similar remanence coercivity spectra and thermal demagnetization characteristics. In these rocks low-coercivity minerals, magnetite and coarse-grained haematite are prevalent but high-coercivity components attributed to haematite pigment are also present (Fig. 2e). The latter was shown to carry a secondary magnetization of invariably normal polarity usually with a wide range of blocking temperatures. Fortunately, the bulk overprint was removable thermally at 450° C (Fig. 4a) and a magnetic polarity sequence could be established for the Pliensbachian stage from the Bakonycsernye section (Márton et al., 1980). It was found, however, that even above 450° C the centres of the normal and reversed clusters were not exactly opposite each other and also the reversed directions exhibited a greater dispersion than the normal ones, i.e. a small normal overprint was still present. Nevertheless, after demagnetization at the optimum temperature, (cf. Table I) the normal directions were no longer affected appreciably and they may be used for the geologically assigned period unlike the reversed directions which could only be cleaned sufficiently in exceptional cases (Fig. 4a).

The white limestone of Tithonian age from Lókut-Hill appears to contain no other magnetic mineral than magnetite (Fig. 2f). The natural remanence of the samples was overprinted in the initial state to varying degrees but it could be isolated after demagnetization at 500° C (Fig. 3b).

Cretaceous localities: Berzsekhegy quarry (*15*) is situated near Lábatlan in the Gerecse Mountains where dark-grey marls of Valanginian—Hauterivian age were sampled. Aptian crinoideal grey limestones were collected at one site each from the Bakony and Vértes Mountains near Borzavár (*16*) and Vértessomlyó (*17*) villages, respectively. The samples from upper Senonian greyish-white marls originate from two quarries at Magyarpolány (*18*) and Bakonyjákó (*19*) villages, both in the western region of the Bakony Mountains some 10 km north of the mining town of Ajka.

In the Berzsekhegy dark-grey marls the predominant magnetic mineral was found to be magnetite (Fig. 2g). Certain samples were slightly overprinted by the present magnetic field but practically a single component was isolated at 400° C. The optimum cleaning temperature was 500° C but most specimens blew up at this temperature.

In the grey crinoideal limestones magnetite and pigmentary haematite were distinguishable from the remanence coercivities (Fig. 2h). The natural remanence isolated after demagnetization at 500° C resides in the magnetite because the remanence directions ceased to be coherent after demagnetization above the Curie-point of magnetite.

The remanence coercivities indicated magnetite for the Senonian samples,

too. The NRM was isolated above 500° C except for extremely weak specimens which could not be measured after demagnetization at temperatures higher than 200° C.

Eocene localities: The samples representing these were collected from white limestones in an abandoned bauxite pit (*20*) near Darvastó and in a roadside quarry (*21*) at Szöc villages, both in the Bakony Mountains but they were too weak for precise measurements.

IDENTIFICATION OF MAGNETIZATION

The magnetic properties of the formations were discussed in the preceding section and it has been shown that magnetite is the chief contributor to the natural remanence. In the red limestones of lower Jurassic age haematite was also found to be an important remanence carrier and it was shown (Márton et al., 1980) that the directions of magnetization associated with magnetite and haematite are parallel. Thus, we may conclude that the directions of the natural remanence represent the Earth's magnetic field at the time of deposition or shortly after.

It has also been shown that a high coercivity overprinting remanence of normal polarity associated with haematite pigment is present in the older Mesozoic formations. This secondary magnetization is definitely pre- (upper) Tertiary because of its characteristic westerly declination and it is believed to be CRM acquired in late diagenetic processes during one of Mesozoic periods of normal polarity. In fact, this is the only characteristic remanence found in the Dachsteinkalk-limestone at Borzavár and in part of the crystalline limestone at Polgárdi. The presence of this remanence rendered AF-demagnetization impracticable and it was only because its blocking temperature is lower than that of the primary remanence that the latter could be isolated by thermal demagnetization.

CRM is not the only secondary magnetization encountered. It has been found that in purely magnetite-bearing rocks (e.g. the Tithonian limestone at Lókut-Hill) relatively high cleaning temperatures (400—500° C) were needed to destroy a secondary remanence which is attributed to VRM.

STABILITY INDICATORS

Tilt (fold) test: The rocks at most sampling sites appear in the form of gently dipping beds of monoclincal attitude, so that a tilt (fold) test within a single site was not feasible in general. However, the samples were taken from either limb of a small asymmetrical anticline at Vértessomlyó (Aptian limestone) and from beds dipping roughly in the same direction but at different angles between 5° and 30° at Magyarpolány (Senonian marl) and the grouping of the NRM directions improved after unfolding, respectively tectonic tilt correction. Similarly, each Triassic, Jurassic and Cretaceous site mean direction of magnetization became more precise after tectonic tilt correction.

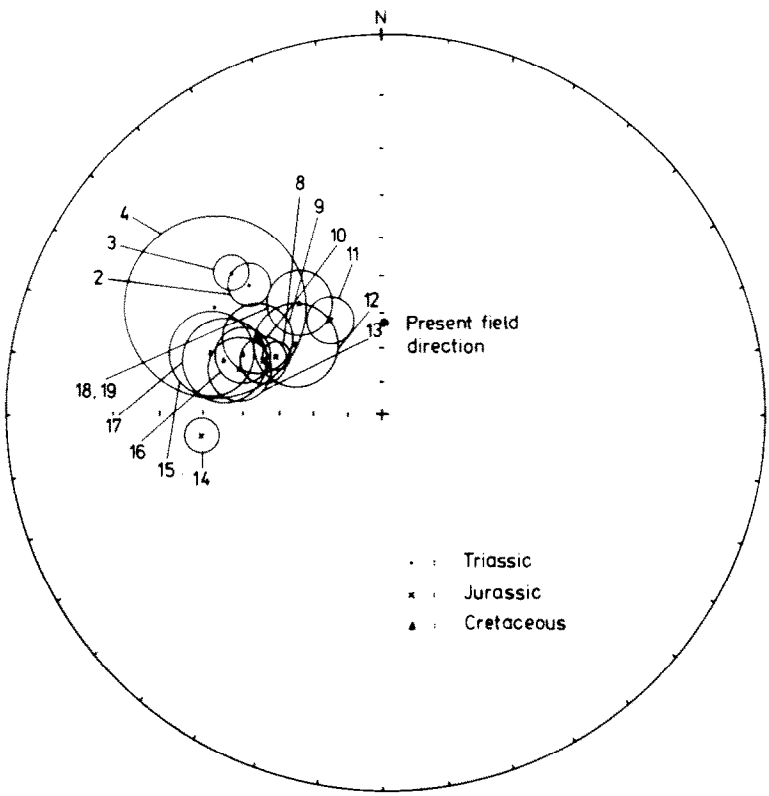


Fig. 5. Stereographic plot of the site mean directions of natural remanence with confidence circles (α_{95}) after tectonic tilt correction for Mesozoic localities from the Transdanubian Central Mountains (Table I).

Polarities: Apart from the Jurassic limestone sections in which normal and reversed polarities occurred in about equal proportions, normal polarities prevailed both in the Triassic and Cretaceous collections. In fact, none of the Cretaceous samples showed reversed magnetization and only one or two of them were reversely magnetized from each Triassic locality.

The presence of the normal and reversed magnetizations may indicate NRM stability in general and so it is interpreted for Triassic and Jurassic sites. But the Cretaceous formations may all have been deposited during normal polarity intervals including the Cretaceous normal period (Irving and Pullaiah, 1976) and with these just the lack of reversed polarity may be indicative of NRM stability.

NRM directions: The tilt-corrected site mean directions are shown in Fig. 5 and we only point out here that they are reasonably precise and all are significantly different from the present field direction which is also valid for the uncorrected NRM directions.

Thus, the stability indicators also support the former conclusion that the thermally cleaned natural remanences are true records of the geomagnetic field at the time of sedimentation.

PALAEOMAGNETIC RESULTS AND THEIR IMPLICATIONS

Table I summarizes the site mean directions of natural remanence after tectonic tilt correction (Fig. 5) and other relevant palaeomagnetic data. To diminish the effect of more complex rotations that may be involved, we computed overall average directions of magnetization for four periods as follows: lower (middle) Triassic, lower Jurassic, lower Cretaceous and upper Senonian. In these calculations we used all samples instead of taking the averages of the locality means which would have resulted in a pessimistic α_{95}. Greater differences in the number of samples were, where applicable, allowed for by combining remanence directions from neighbouring strata into a single determination. The final number of "samples" (individual or combined) from each locality is also indicated in Table I. The Tithonian result (and probably the Devonian, too) seems to be anomalous because of the large westerly palaeodeclinations. Either being a single determination it is possible that the tilt-corrected remanence directions are still affected by some local rotational tectonics.

The palaeopole (VGP) positions are shown in Fig. 6 and the apparent polar wander (APW) curves for (stable) Europe and Africa are also included for comparison. Both curves were derived from Irving's APW path for North America (1977), utilizing the finite rotations of North America with respect to Europe and Africa in Sclater et al., (1977).

Recently VandenBerg (1979) has reviewed all available APW curves for Africa and arrived at the conclusion that the middle and late Mesozoic and Tertiary parts of the indirect polar wander curves by Irving (1977) and by Van der Voo and French (1974) are in good agreement with the observed palaeomagnetic poles of Africa (cf. table III in VandenBerg, 1979). He also pointed out that the "African" curve obtained originally by him from Irving's APW path for North America was very similar to the inferred polar wander curve of Van der Voo and French (1974). According the VandenBerg, the APW curve for North America is fairly well specified and the African data (direct and indirect) seem to confirm this statement. So, if it is correct to assume that North America and Europe had not been in relative motion during Mesozoic times before they began to separate 95 m.y. ago (Sclater et al., 1977) then the transferred section of the "American" curve in Fig. 6 is certainly a better choice for the "European" APW than that which could be constructed from the very scarce direct observations for the Western-European Mesozoic.

The APW path for the Transdanubian Central Mts. clearly resembles the "African" curve (Fig. 6) in that the Mesozoic sections of both APW curves describe a loop which is not present in the "European" polar wandering.

TABLE I

Summary of palaeomagnetic results from Mesozoic rocks of the Transdanubian Central Mountains (mean geographic coordinates are $\phi = 47.5°$; $\lambda = 18.1°$). All computations were carried out following Fisher (1953).

Locality	Number of samples	Magnetization			Palaeomagnetic pole				Cleaning temperature (°C)
		Decl. (°)	Incl. (°)	α_{95} (°)	Lat. (°)	Long. (°)	Δp (°)	Δm (°)	
2 Száka-Hill	12 (6)	315	38	5.0	46	296	3.5	5.9	500
3 Csopak, Quarry I.	9 (9)	314	33	4.0	42	266	2.6	4.5	500
4 Csopak, Quarry II.	5 (5)	303	35	23.5	37	277	15.7	27.2	200
Triassic l (m) mean	20 *	311	35	5.3	42	270	3.5	6.1	
8 Bakonycsernye	54 (9)	299	55	2.6	44	297	2.6	3.7	450
9 Bakonycsernye	15 (8)	295	52	5.8	39	296	5.4	7.9	450
10 Bakonycsernye	7 (7)	302	48	8.9	42	287	7.5	11.6	450
11 Tata	10 (10)	332	58	5.7	68	275	6.2	8.4	500
12 Tardosbánya	6 (6)	310	57	10.7	52	291	11.2	15.5	400
13 Lókut	21 (7)	288	47	8.0	32	296	6.6	10.3	460
Jurassic l (m) mean	47 *	304	54	4.0	47	292	3.9	5.6	
Tithonian									
14 Lókut	14 (14)	264	39	4.1	12	307	2.9	4.9	500
15 Berzsekhegy	10 (10)	290	39	9.9	29	290	7.0	11.8	500 (400)
16 Borzavár	11 (11)	294	47	7.4	37	292	6.2	9.6	500
17 Vértessomlyó	7 (7)	285	47	10.4	30	298	8.6	13.4	500
Cretaceous (*l*) mean	28 *	290	43	4.9	32	292	3.8	6.2	
Senonian									
18 Magyarpolány	5 (5)	324	50	8.6	58	270	7.7	11.5	500
19 Bakonyjákó	2 (2)								200

* Combined, see text

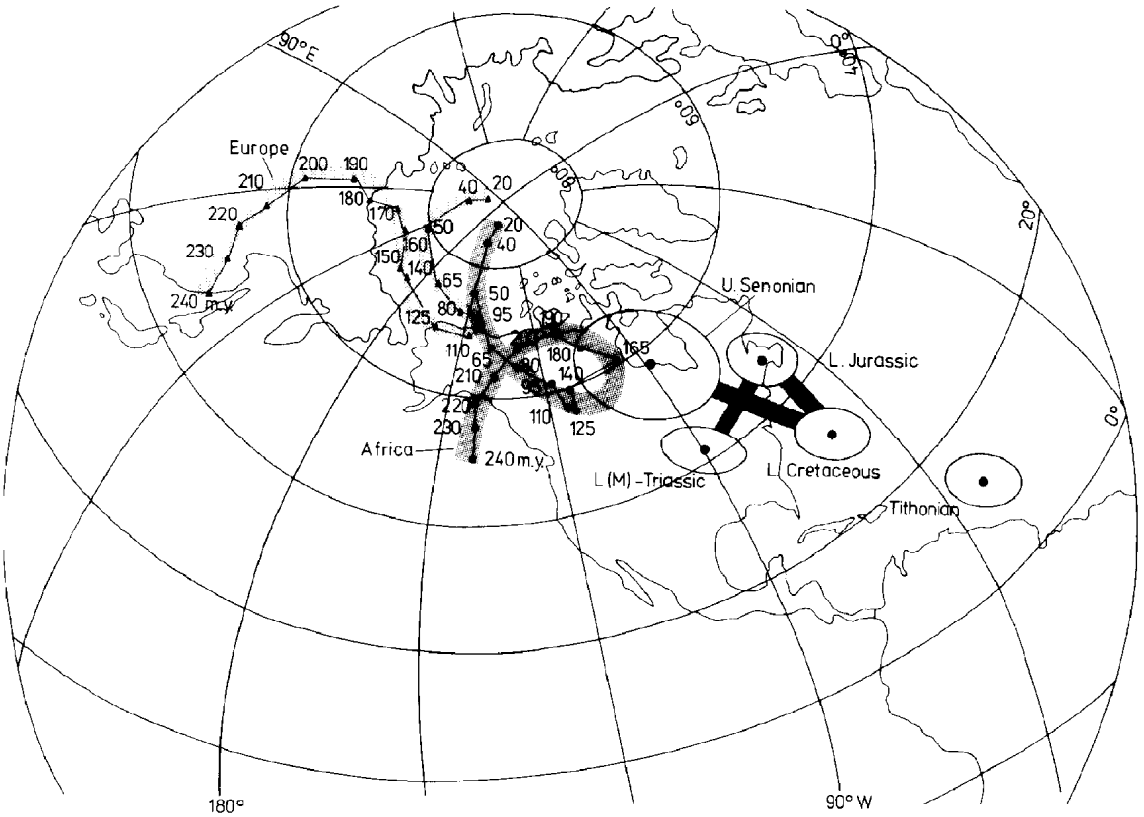


Fig. 6. Mesozoic palaeomagnetic pole positions of the Transdanubian Central Mountains with α_{95}-confidence ovals and apparent polar wander (APW) curves for stable Europe and Africa. Both curves were recomputed from the APW path of North America (Irving, 1977) using the finite rotations of the latter relative to Europe and Africa (Sclater et al., 1977). (Oblique azimuthal equidistant projection).

Furthermore, it can be seen that although the palaeomagnetic poles for the Central Mountains fall on a distinctly different region and away from the polar paths of both Africa and Europe this region is nearer to the "African" and farther away from the "European" palaeomagnetic poles. Thus, it is quite obvious that the palaeomagnetic results for the Central Mountains have an African affinity but the existing difference between the respective polar wanderings needs explanation. However, all four Mesozoic palaeomagnetic poles of the Transdanubian Central Mountains can be brought practically into the corresponding "African" poles the simplest by a 35° clockwise rotation of the measured palaeodeclinations which means that the present difference between the Mesozoic poles of the Central Mountains and Africa is explicable by a counterclockwise rotation of the Central Mountains after upper Cretaceous times. This movement is similar to the Tertiary rotation of

the Adriatic block relative to Africa (VandenBerg, 1979) and they may be in causal relationship.

Concluding, our palaeomagnetic results give independent evidence that the Transdanubian Central Mountains originated from the southern margin of the Mesozoic Tethys as has been suggested earlier on palaeontological grounds (Géczy, 1973). Also the Central Mountains may have been part of the African plate throughout Mesozoic times and the 35° counterclockwise declination rotation is interpreted as the palaeomagnetic signature of the Tertiary emplacement of the Central Mountains into their present geographic position.

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